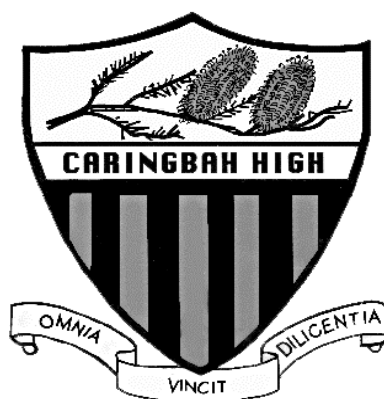




Caringbah High School

Year 11 2016

Mathematics

Preliminary HSC Course

Semester 2 Exam

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A reference sheet with mathematical formulae is provided.
- In Questions 6 - 12, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for partial or incomplete answers

Total marks – 85

Section I 5 marks

Attempt Questions 1-5
Mark your answers on the answer sheet provided. You may detach the sheet and write your name on it.

Section II 80 marks

Attempt Questions 6 - 12
Write your answers on the answer sheets provided. Ensure your name or student number is clearly visible.

Name: _____

Class: _____

<i>Marker's Use Only</i>									
Section I	Section II							Total	
Q 1-5	Q6	Q7	Q8	Q9	Q10	Q11	Q12		
/5	/12	/12	/12	/11	/11	/11	/11	/85	%

Question 6 (12 marks) Start a NEW page.

Marks

- a) Determine the midpoint of the interval AB with $A (-1, 3)$ and $B (11, -6)$. **1**
- b) Rationalise the denominator and simplify $\frac{2\sqrt{3}}{2\sqrt{2} + 3}$ **2**
- c) If $3x - ky + 7 = 0$ is parallel to $y = 4x - 5$, find the value of k . **2**
- d) Simplify $\frac{\cos(90^\circ - \theta)}{\cos \theta}$ **2**
- e) Simplify $2\sqrt{54} + \sqrt{150}$ **2**
- f) Find the equation of the line that is perpendicular to $5x + 2y - 4 = 0$ that also passes through the point $(-1, 2)$. Write your answer in general form. **3**

Question 7 (12 marks) Start a NEW page.

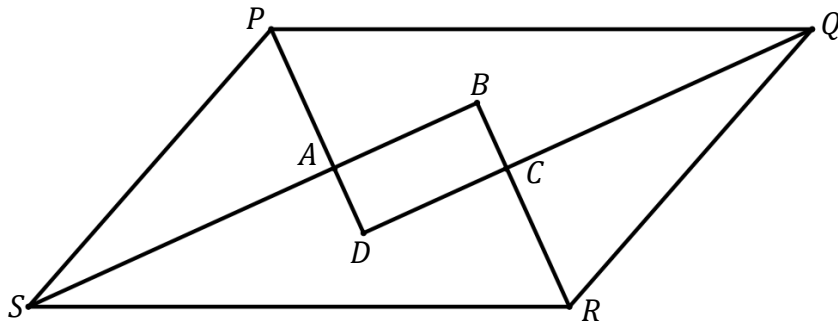
Marks

- a) If $f(x) = 1 - x$, write a simplified expression for $f(1 - x)$

1

- b) The quadrilateral $PQRS$ is a parallelogram with $SR \neq RQ$.
The bisectors of its angles intersect at A, B, C and D as shown.

3



Copy or trace the diagram onto your answer page and prove that quadrilateral $ABCD$ is a rectangle.

- c) i) Sketch the following piecemeal function in the domain $-2 \leq x \leq 6$

3

$$f(x) = \begin{cases} -3 & \text{for } x < -1 \\ -x^2 - 2 & \text{for } -1 \leq x \leq 2 \\ x - 8 & \text{for } x > 2 \end{cases}$$

- ii) What is the range of $f(x)$ for the given domain?

1

- d) Solve $x^2 - x - 2 > 0$

2

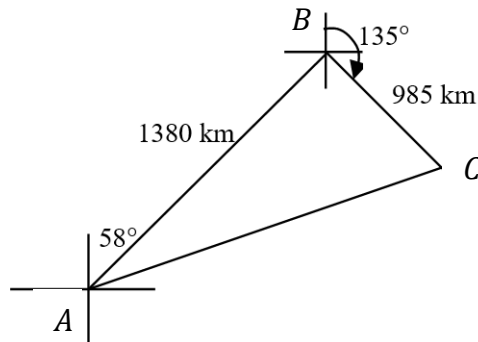
- e) Find the domain and range of

$$y = \frac{1}{\sqrt{4 - x^2}}$$

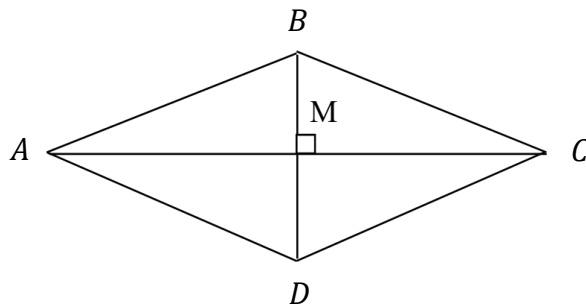
2

Question 8 (12 marks) Start a NEW page.**Marks**

- a) A plane travels 1380 km from A to B on a bearing of 58°T .
The plane then travels 985 km on a bearing of 135°T to a point C .



- | | | |
|------|---|----------|
| i) | Show that $\angle ABC$ is 103° . | 1 |
| ii) | Find the distance of CA , correct to one decimal place. | 2 |
| iii) | Find the size of $\angle BAC$, to the nearest degree. | 2 |
| iv) | Find the bearing of A from C , to the nearest degree. | 1 |
- b) The area of rhombus $ABCD$ is 80 cm^2 . One diagonal is $2\frac{1}{2}$ times as long as the other.



- | | | |
|-----|---|----------|
| i) | Find the length of the shorter diagonal. | 2 |
| ii) | Find the length of the sides of the rhombus | 2 |
- c) Evaluate $\lim_{x \rightarrow -4} \frac{x^2 + 3x - 4}{x + 4}$
- 2**

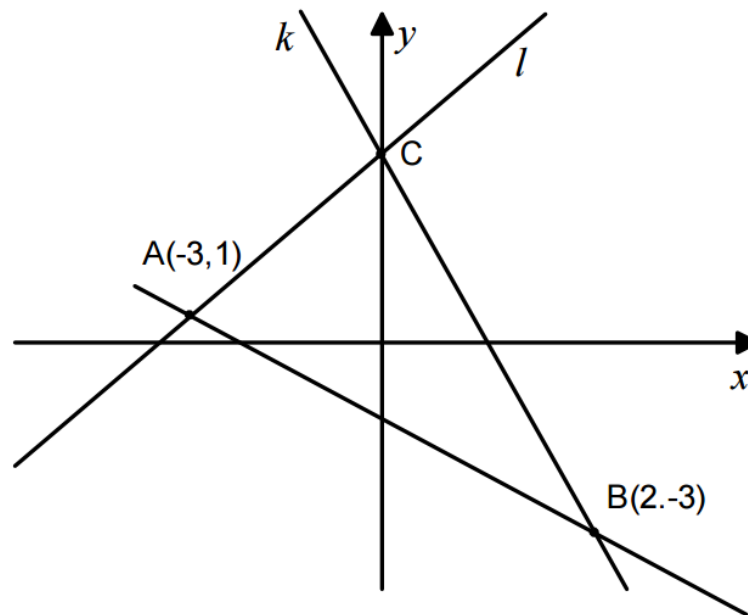
Question 9 (11 marks) Start a NEW page.**Marks**

a) Prove that $\sec^2 \theta - 2 \tan \theta = (\tan \theta - 1)^2$. 2

b) Given points $A(-3, 1)$ and $B(2, -3)$.

Point A lies on the line l given by the equation $4x - 3y + 15 = 0$ and

Point B lies on the line k given by the equation $4x + y - 5 = 0$.



The questions below refer to the diagram drawn above.

Copy the diagram onto your answer page.

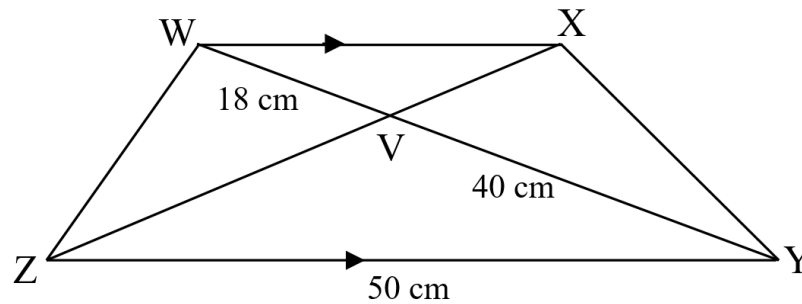
- | | | |
|-------|---|---|
| (i) | Show that the point C , the point of intersection of the lines l and k , must lie on the y -axis. | 2 |
| (ii) | Find the gradient of the line AB . | 1 |
| (iii) | Show the equation of the line AB is $4x + 5y + 7 = 0$. | 2 |
| (iv) | Find the perpendicular distance from point C to the line AB . | 2 |
| (v) | Find the exact area of $\triangle ABC$. | 2 |

Question 10 (11 marks) Start a NEW page.

Marks

a) Solve $|x - 4| = 2x + 1$. 3

b) $WXYZ$ is a trapezium with $WX \parallel ZY$. $WV = 18$ cm, $VY = 40$ cm and $ZY = 50$ cm.



i) Show that $\triangle WXV \parallel \triangle YZV$, giving reasons. 3

ii) Find the length of WX . 2

c) Shade the region given by $y > x^2 - 1$ and $y \leq 3^x$ 3

Question 11 (11 marks) Start a NEW page.

Marks

a) i) Given that a circle has an equation $x^2 + 4x + y^2 - 6y - 36 = 0$, find the centre and radius of the circle. 2

ii) Shade the region defined by $x^2 + 4x + y^2 - 6y - 36 \leq 0$ 2

b) Find the value of x if $2 \times (2^{2x}) - 2^x = 0$. 3

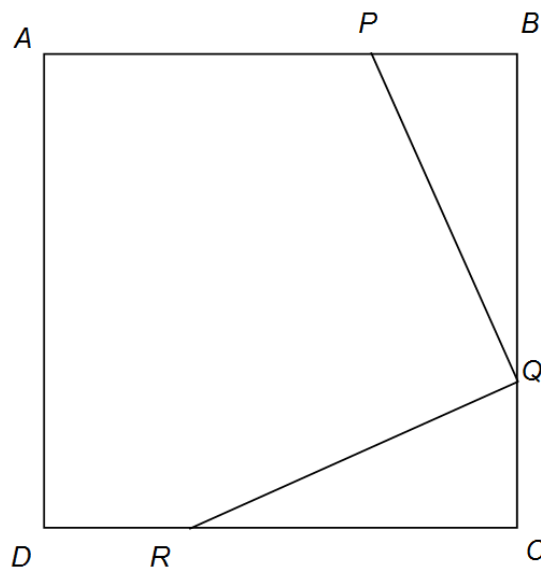
c) i) On the same number plane sketch the graphs $y = x^2 - 4$ and $y = 4x + 1$. 2

ii) Hence or otherwise, find the points of intersection of these graphs. 2

Question 12 (11 marks) Start a NEW page.

Marks

- a) Factorise fully $-192x^3 - 3$ 2
- b) Show that the line $3x + 4y - 12 = 0$ does not intersect the circle $x^2 + y^2 = 4$ 3
- c) $ABCD$ is a square P, Q and R are points on AB, BC and CD respectively such that $AP = BQ = CR$.



- i) Prove that $\triangle PBQ \cong \triangle QCR$ 3
- ii) Hence or otherwise, prove that $\angle PQR = 90^\circ$. 3

END OF EXAM

Name: _____

Multiple choice answer page. Fill in either A, B, C or D for questions 1-5.

This page must be handed in with your answer booklets

1.	
2.	
3.	
4.	
5.	

Year 11 2016 Mathematics Preliminary HSC Course Semester 2 Exam Answers

Q1) $f(a) = a^2 - a + 1 = 3$
 $\therefore a^2 - a - 2 = 0$
 $\therefore (a - 2)(a + 1) = 0$
 $\therefore a = -1 \text{ or } 2$
 (C)

Q2) $y = \frac{1}{x}$ is an odd function (A)

Q3) $2x - 4y - 3 = 0$
 $\therefore y = \frac{1}{2}x - \frac{3}{2}$
 $\therefore m = \frac{1}{2}, b = -\frac{3}{2}$ (B)

Q4) $\sin(270^\circ + \theta)$
 $= \sin(360^\circ - (90^\circ - \theta))$
 $= -\sin(90^\circ - \theta)$
 $= -\cos \theta$ (D)

Q5) **Exterior Angle** $= 360^\circ / 18 = 20^\circ$
 $\therefore$ **Interior Angle** $= 180^\circ - 20^\circ$
 $\therefore$ **Interior Angle** $= 160^\circ$ (B)

Q6) a) **Midpoint** $= \left(\frac{(-1) + 11}{2}, \frac{3 + (-6)}{2} \right)$
 $\therefore$ **Midpoint** $= \left(5, -\frac{3}{2} \right)$

b) $\frac{2\sqrt{3}}{2\sqrt{2} + 3} \times \frac{2\sqrt{2} - 3}{2\sqrt{2} - 3} = \frac{4\sqrt{6} - 6\sqrt{3}}{-1}$
 $= -4\sqrt{6} + 6\sqrt{3}$

c) $y = 4x - 5 \therefore m = 4$
 $3x - ky + 7 = 0 \therefore y = \frac{3}{k}x + \frac{7}{k}$
 $\therefore \frac{3}{k} = 4 \therefore k = \frac{3}{4}$

d) $\frac{\cos(90^\circ - \theta)}{\cos \theta} = \frac{\sin \theta}{\cos \theta} = \tan \theta$

e) $2\sqrt{54} + \sqrt{150} = 6\sqrt{6} + 5\sqrt{6}$
 $= 11\sqrt{6}$

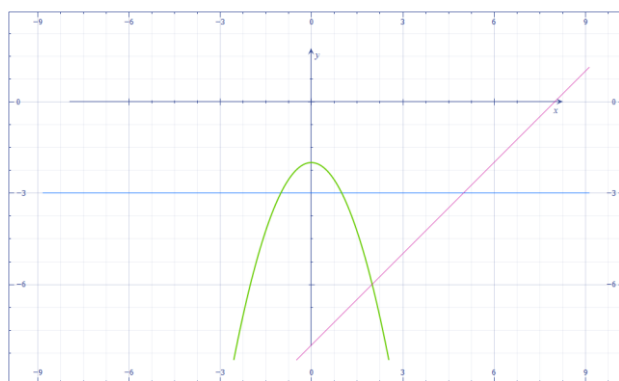
f) $5x + 2y - 4 = 0 \therefore y = -\frac{5}{2}x + 2$
 $\therefore m = -\frac{5}{2}, \therefore m_\perp = \frac{2}{5}$
 $\therefore$ line through $(-1, 2)$
 $y - 2 = \frac{2}{5}x - (-1)$
 $\therefore y = \frac{2}{5}x + 3$

Q7)

a) $f(1 - x) = 1 - (1 - x) = x$

b) Let $\theta = \angle PSA = \angle ASR$ (bisector of $\angle PSR$)
 $= \angle PQC = \angle CQR$ (bisector of $\angle PQR$)
 $\therefore \angle SPA + \angle APQ = 180^\circ - 2\theta$ (co-int PQ//SR)
 $\therefore \angle SPA = \angle APQ = 90^\circ - \theta$ (bisectors of $\angle SPQ$)
 $\therefore \angle PDQ = 180^\circ - (\angle APQ + \angle PQC)$ (sum $\triangle PDQ$)
 $\therefore \angle PDQ = 90^\circ$
 $\therefore$ similar for $\angle SBR = 90^\circ \therefore$ rectangle

c) i)



ii) Range $y \geq -6$

d) $x^2 - x - 2 > 0$
 $\therefore (x - 2)(x + 1) > 0$
 $\therefore -1 < x < 2$

e) Domain $-2 < x < 2$, Range $y \geq \frac{1}{2}$

Q8)

a) i) $\angle ABC = 56 + 45 = 103^\circ$
 ii) $CA^2 = 1380^2 + 985^2 - 2 \times 1380 \times 985 \times \cos 103$
 $\therefore CA = 1867.1 \text{ km}$
 iii) $\frac{\sin A}{985} = \frac{\sin 103}{1867.1}, \therefore A = 30^\circ 56' \approx 31^\circ$
 iv) 269° T

b) i) $A = \frac{1}{2}xy, \therefore 80 = \frac{1}{2} \times x \times \frac{5}{2}x$
 $\therefore 80 = \frac{5}{4}x^2, \therefore x = 8$
 ii) $AB = \sqrt{4^2 + 10^2} = \sqrt{116}$

c) $\lim_{x \rightarrow -4} \frac{x^2 + 3x - 4}{x + 4} = \lim_{x \rightarrow -4} \frac{(x + 4)(x - 1)}{x + 4}$
 $= \lim_{x \rightarrow -4} (x - 1) = -5$

Q9)

a) $RHS = (\tan \theta - 1)^2 = \tan^2 \theta - 2 \tan \theta + 1$
 $= (\sec^2 \theta - 1) - 2 \tan \theta + 1$
 $= \sec^2 \theta - 2 \tan \theta = LHS$

b) i) $l: 4x - 3y + 15 = 0$ $k: 4x + y - 5 = 0$

$\therefore 4x - 3y + 15 = 0 \quad - \textcircled{1}$

$\therefore 4x + y - 5 = 0 \quad - \textcircled{2}$

$\therefore \textcircled{1} - \textcircled{2} \quad -4y + 20 = 0$

$\therefore y = 5, x = 0 \therefore C$ is on y - axis

ii) $m = -\frac{4}{5}$

iii) $y - 1 = -\frac{4}{5}(x + 3)$

$\therefore 5y - 5 = -4x - 12$

$\therefore 4x + 5y + 7 = 0$

iv) $d_{\perp} = \frac{|4 \times 0 + 5 \times 5 + 7|}{\sqrt{4^2 + 5^2}} = \frac{32}{\sqrt{41}}$

v) $AB = \sqrt{5^2 + 4^2} = \sqrt{41}$

$Area = \frac{1}{2} \times \sqrt{41} \times \frac{32}{\sqrt{41}} = 16 (u^2)$

Q10)

a) $x - 4 = 2x + 1$ or $-x + 4 = 2x + 1$
 $\therefore x = -5$ or $x = 1$

b) i) In the Δ 's WXY & YZV

$\angle XWV = \angle VYZ$ (alt $\angle$'s, $WX \parallel YZ$)

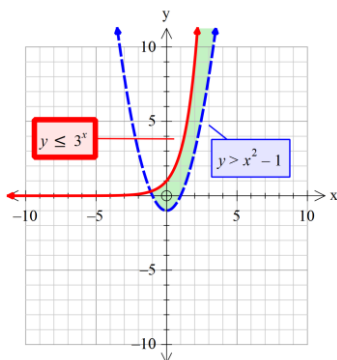
$\angle WVX = \angle YVZ$ (vert opp.)

$\angle WXV = \angle YZV$ ($\angle$'s in Δ)

$\therefore \Delta WXV \parallel \Delta YZV$

ii) $\frac{WX}{50} = \frac{18}{40} \therefore WX = 22.5$

c)



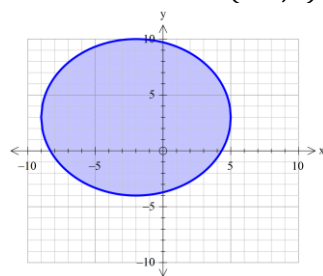
Q11)

a) i) $x^2 + 4x + 4 + y^2 - 6y + 9 = 36 + 4 + 9$

$\therefore (x + 2)^2 + (y - 3)^2 = 49$

$\therefore$ Circle: Centre $(-2, 3)$ Radius = 7

ii)



b) Let $u = 2^x, \therefore 2u^2 - u = 0$

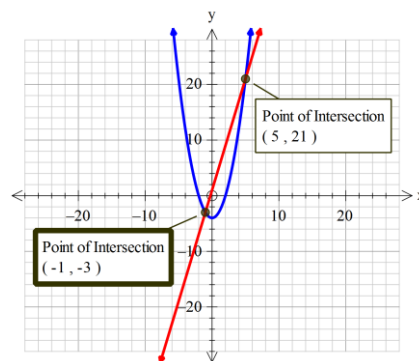
$\therefore u(2u - 1) = 0$

$\therefore u = 0$ or $\frac{1}{2}, \therefore 2^x = 0$ or $\frac{1}{2},$

But $2^x = 0$ has no solutions.

$\therefore x = -1$

c) i)



ii) $y = x^2 - 4$ and $y = 4x + 1$

$\therefore x^2 - 4 = 4x + 1$

$\therefore x^2 - 4x - 5 = 0$

$\therefore (x + 1)(x - 5) = 0$

$\therefore x = -1$ or 5

$\therefore (-1, -3)$ or $(5, 21)$

Q12)

a) $-192x^3 - 3 = -3(64x^3 + 1)$
 $= -3(4x + 1)(16x^2 - 4x + 1)$

b) Circle $x^2 + y^2 = 4 \therefore$

centre $(0, 0)$ radius 2

But perpendicular distance from $(0, 0)$ to the line $3x + 4y - 12 = 0$

$d_{\perp} = \frac{|3 \times 0 + 4 \times 0 - 12|}{\sqrt{3^2 + 4^2}} = \frac{12}{5} > 2$

$\therefore$ no point(s) of intersection

c) i) Since $AP = BQ = CR$ then $BP = QC = DR$

In the Δ 's PBQ & QCR

$PB = QC$ (above)

$\angle PBQ = \angle QCR$ (prop. of square)

$BP = CR$ (given)

$\therefore \Delta PQB \equiv \Delta QCR$ (SAS)

ii) Let $\theta = \angle BQP$

$\therefore \angle BPQ = 90 - \theta$

($\angle$'s in Δ since $\angle PBQ = 90$)

$\therefore \angle RQC = 90 - \theta$

(corr. $\angle$'s in cong. Δ 's)

$\therefore \angle CQR + \angle RQP + \angle BQP = 180$

$\therefore \angle CQR = 180 - (90 - \theta) - \theta$

$\therefore \angle CQR = 90^\circ$